MALL-ACC-08 Malleswaram Node Access Role Huawei Optical Transport Failure SOP

Node: MALL-ACC-08 | **Vendor:** Huawei | **Severity:** Warning | **Alarm Code:** ALARM_2711

Description

The Huawei Access equipment at Malleswaram Node has been afflicted by an Optical Transport Failure, specifically an ALARM_2711 error condition, indicating a Warning severity. This failure domain is rooted in the degradation of the SFP/XFP, a phenomenon that may be precipitated by the wear and tear on the optical fibers, leading to a decrease in the Receive (Rx) or Transmit (Tx) power. The underlying protocol mechanics of the Optical Transport Network (OTN) are predicated on the reliable transmission of optical signals over fiber optic cables, which necessitates a stable and optimal signal-to-noise ratio. When the SFP/XFP degrades, it may lead to packet loss, errors, or retransmissions, ultimately resulting in an ALARM_2711 condition. In addition, physical cuts to the fiber optic cables or other environmental factors may also contribute to this failure domain. A comprehensive understanding of the OTN protocol and the Huawei Access equipment's architecture is essential to effectively diagnose and resolve this issue.

Pre-requisite Checks

- Check: Verify the status of the redundant link to ensure it is operational and can take over in case of a primary link failure.

- Check: Dispatch remote hands to the Malleswaram Node to physically inspect the equipment and fiber optic cables.

Potential Root Causes

Cause	Probability
SFP/XFP degradation	High
Fiber cuts	Medium
Rx/Tx power loss	Low

Diagnostic Commands & Expected Outputs

```
Command: display fault brief, Expected Output: Fault Brief Information, Description: This command displays a brief summary of the fault information, including the alarm code and severity. It helps to quickly identify the root cause of the issue.

Command: display fault detail, Expected Output: Fault Detail Information, Description: This command provides a detailed view of the fault information, including the fault type, location, and timestamp. It is essential to understand the fault details to determine the root cause.
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Command: display optical interface, Expected Output: Optical Interface Information, Description: This command displays the optical interface information, including the transmit and receive power levels. It helps to identify any issues related to the SFP/XFP or fiber optic cables.
```

Resolution Steps

Step 1: Step: Replace the SFP/XFP, Command: replace sfp/xfp , Description: Replace the degraded SFP/XFP with a new one. This step requires physical access to the equipment and fiber optic cables.

Step 2: Step: Repair or replace the fiber optic cable, Command: repair/replace fiber optic cable , Description: Repair or replace the damaged fiber optic cable. This step requires physical access to the equipment and fiber optic cables.

Step 3: Step: Adjust the Rx/Tx power levels, Command: adjust receive/transmit power , Description: Adjust the receive and transmit power levels to ensure optimal signal-to-noise ratio. This step requires a deep understanding of the OTN protocol and the Huawei Access equipment's architecture.

Post-Check Verification

```
Command: display fault brief, Expected Output: Fault Brief Information, Description: Verify that the alarm code and severity have been cleared.

Command: display optical interface, Expected Output: Optical Interface Information, Description: Verify that the optical interface information, including the transmit and receive power levels, is within the normal operating range.
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Roll-back Procedures

- Step: Restore the original SFP/XFP, Command: restore sfp/xfp , Description: Restore the original SFP/XFP if the replacement fails.

- Step: Restore the original fiber optic cable, Command: restore fiber optic cable , Description: Restore the original fiber optic cable if the replacement fails.